

Cost-Reduction Suggestions

MASMEC budget quote C610222-1 vs. Modular All Drives Test System Definitions (rev.00)

Objective: reduce the €1,250,000 base price while preserving the concept's modularity and scalability. Estimates are engineering judgement — MASMEC priced per module, not per component, so figures are indicative and require a MASMEC re-quote to confirm.

Guiding principle: the **Load module** is the unit you replicate to scale (20-min cycle = the bottleneck). Every euro removed from the load module multiplies across the line, so the highest-value savings are the ones that lower the **per-load-module** cost.

1. Items you already identified — confirmed

- **Manual load/unload stations (Opt.1, OP10-M / OP50-M): remove — save €150,000.** Your concept already loads, wires and QR-reads the product on the fixture at the assembly stage (operational flow steps 1–2), and returns fixtures on the lower conveyor. The stations duplicate this. *Caveat (MASMEC):* add one fixed QR reader upstream of the hipot module (~€5–10k) so pallet RFID can be written without OP10-M. Net saving ≈ €140k.
- **HMI button robot (OP30A F04, DENSO 4-axis): remove — save ~€30–50k.** Replace with a low-cost **per-fixture pneumatic pusher** aligned to each model's button layout, or validate the keypad via a **fieldbus/serial keypad self-test** in firmware. Keep the HMI vision camera for display verification.

2. Concept improvements (industry-aligned) to cut price

2.1 Replace programmable grid simulators with tap-switched transformers

Both OP30A and OP40A include a Siemens SINAMICS S120 active grid simulator (220/440/575 V, 60 A regenerative) purely to derive the voltage classes from your 380 V plant supply. For 100%-of-units end-of-line testing, worldwide VFD lines use a **multi-tap / auto-transformer with contactor tap selection** for voltage classes and reserve programmable AC sources for **type testing** (a few units), not production. **Indicative saving €60–120k per module that uses it.** Keep a programmable source only if an EOL test genuinely requires variable voltage/frequency or grid-disturbance (sag/swell) injection.

2.2 Production-grade instrumentation, not lab-grade

The Load module specs a Yokogawa WT5000 (0.03%-class) + 7 input modules + 7 probes. EOL testing is go/no-go against limits — **LEM transducers + DAQ** (exactly what MASMEC already proposed for the Functional station, where it states 'no power analyzer') are sufficient. **Indicative saving €20–40k per load module.** Keep at most **one** high-accuracy reference analyzer, shared across the line for audit/calibration — not one per replicated module.

2.3 Defer undefined scope & phase fixtures

- **'Additional vision check' camera (OP30A F06, TBD):** defer until a real inspection need is defined — save ~€5–10k.
- **Changeover fixtures (5 types × 3):** buy each type as its product model launches rather than all up front (Opt.3B €4k/set) — defers capex and matches the pay-as-you-grow scaling logic.

2.4 Server & standardization

- **Server:** host the database on existing WEG IT / MES and buy only the data-collection/DataRelay software — save ~€10–25k and improve plant-system integration.
- **Common module 'shell':** standardize frame, safety, conveyor segment, PLC and HMI across all module types (MASMEC already shares F01–F03) so each replicated load module is cheaper to build and stock spares for — a direct scalability lever.

2.5 Optional — merge HIPOT + Functional into one station

Hipot (<1 min) + functional (<5 min) = ~6 min, far below the 20-min load cycle, so neither is ever the bottleneck and neither is ever replicated. Combining them into one cabin (shared conveyor/PLC/HMI/safety) could save

~€80–120k with **no impact on scalability**. **Trade-off:** co-locating 4 kV hipot with functional electronics/cameras adds interlock complexity and locks the whole station during HV — a modest reduction in 'one test per module' purity. Decide based on how strictly you hold that rule.

3. Summary of levers

#	Lever	Indicative saving	Modularity / scalability
1	Remove manual load/unload (Opt.1)	€150k (~€8k QR)	Preserved
2	Remove HMI button robot	€30–50k	Preserved
3	Tap transformer vs grid simulator	€60–120k per module	Preserved (self-contained)
4	Production instrumentation vs WT5000	€20–40k per load module	Preserved
5	Defer 'additional' camera	€5–10k	Preserved
6	Server on existing IT/MES	€10–25k	Improved integration
7	Optional: merge HIPOT+Functional	€80–120k	Scalability OK; modularity ↓ slightly
8	Phase changeover fixtures	Defers capex	Pay-as-you-grow
9	Common module shell	Lower per-unit at scale	Improves scalability

Indicative total: applying levers 1–6 removes roughly **€350–450k (~30%)** from the €1.25M base. More importantly, levers 3, 4 and 9 cut the **per-load-module** cost by ~€80–160k each — the saving that compounds as you add load modules toward the ~900/day target.

4. Deliberately kept & a note on WEG's own budget

- **Keep — automated fixture-to-station connection** (pneumatic + match plate): core to your 'automated electrical connection' requirement; your concept already does manual wiring at assembly, so this is the right place to spend.
- **Keep — shared conveyor loop:** MASMEC's centralized loop is cheaper than the concept's 'each module its own conveyor'. Keep it, but **pre-size the loop length now** for future load modules (MASMEC's layout-flexibility note).
- **Keep — STO / ground-fault / short-circuit / comms tests:** safety-critical coverage.
- **WEG-supplied electronic loads:** spec a **small load for the Functional module** (brief motor-operation/current check only) and full 60 A loads only for the Load modules; consider **regenerative loads** to cut the energy/cooling cost of 20-min load tests running across many modules at scale.

Indicative estimates for internal decision support; confirm with a MASMEC re-quote and validate any reduced test coverage (e.g. HMI button method, instrumentation accuracy) against WEG quality requirements before committing.